

# Skills Summary

## Graphing Proportional Relationships

*Use this reference while completing your worksheet or when studying.*

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**The test for proportional.** The graph is a *straight line* and it passes through the *origin* (0,0). Both halves are needed: a straight line that misses the origin is linear but not proportional.

**The constant of proportionality (unit rate).**  $k = \frac{y}{x}$  at any point of the line except the origin. Equivalently  $k$  is the rise over the run.

**The equation.**  $y = kx$  — no added constant, because the line starts at (0,0). Reading a point off the graph gives  $k$ ; the equation then answers questions the drawn axis is too short for.

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### 1. From a table to a graph through the origin

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Plot each row as a point  $(x, y)$  and rule one straight line through them. If the relationship is proportional the line hits (0,0), so (0,0) is always a legal extra point to plot. The unit rate is then the  $y$ -value at  $x = 1$ .

**Example:** A table gives 3 hours of work paying 12 dollars. Plot it with the origin and find the unit rate.

**Step 1.** Pay with no work done is 0 dollars, so (0,0) belongs on the graph; with (3,12) that is enough to rule the line and read its steepness.

**Step 2.**  $k = \frac{12 - 0}{3 - 0} = \frac{12}{3}$ . The line through the origin rises 4 dollars for every hour.

**Try it:** A table gives 4 kilograms of apples costing 10 dollars. What is the cost of one kilogram, in dollars?

check: 2.5

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### 2. Reading the unit rate off a drawn graph

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Find a point where the line crosses a corner of the grid, then divide: the  $y$ -axis quantity goes on top. Dividing the wrong way round gives the reciprocal — a small number where a large one belongs.

**Example:** A line through the origin passes through (6,9). Find its unit rate.

**Step 1.** The line already passes through the origin, so one clean lattice point is all the information a unit rate needs.

**Step 2.**  $k = \frac{9}{6}$ , which is the rise 9 over the run 6. The unit rate is 1.5 per unit of  $x$ .

**Try it:** A different line through the origin passes through (8,28). What is its unit rate?

check: 3.5

### 3. Using the equation to predict

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Once  $k$  is known,  $y = kx$  answers both directions: substitute an  $x$  to predict a  $y$ , or set  $y$  to a target value and solve for  $x$ . This is how you answer a question whose numbers run past the end of the drawn axis.

**Example:** A graph through the origin has unit rate  $k = 4$ . Predict  $y$  when  $x = 7$ , a value past the right-hand end of the drawn axis.

**Step 1.** The graph stops before  $x = 7$ , so read  $k$  off the graph and let the equation  $y = kx$  carry the prediction.

**Step 2.**  $y = 4 \times 7$ . The predicted output is **28**.

**Try it:** A line through the origin has unit rate 6. Solve  $6x = 45$  to find the  $x$  that gives an output of 45.

check: 2.5

**(!) Watch out:** a straight line is not enough. Check the origin before calling a relationship proportional, and check that every  $y \div x$  in a table gives the same number. Dividing  $x$  by  $y$  is the other frequent slip: the unit rate answers “how much  $y$  per one  $x$ ”, so  $y$  is on top.

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